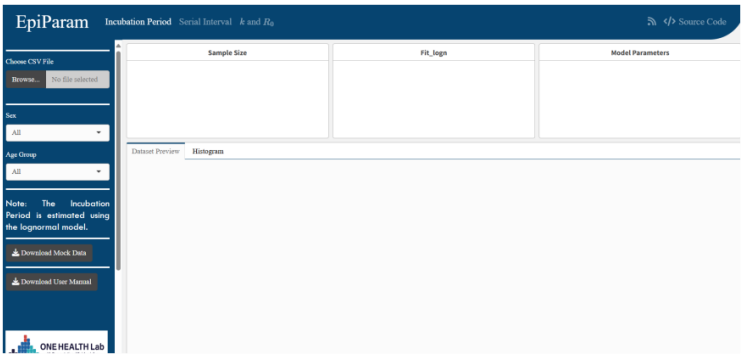
# English Version

# Introduction

EpiParam (Epidemiological Parameter Estimation for Infectious Diseases) is a Shiny interface for estimating five main parameters (incubation period, serial interval, R0, Rt, and k) in the epidemiology of infectious diseases, developed by the OneHealth laboratory of the Department of Preventive Medicine at The Catholic University of Korea, Seoul, Korea.

This user's guide provides instructions on how to use the EpiParam application to generate the incubation period, serial interval, reproduction number (R0), and the Kappa value (K) for a given infectious disease.

To start, paste the link: https://achangwa.shinyapps.io/EpiParam/ into any browser of your choice and run.




This application was built to estimate five parameters: The incubation period, serial interval, reproduction number ($R_0$), and the Kappa value (K).

## Incubation period

This is the period between exposure/infection of an individual by a pathogen and the manifestation of the signs and symptoms of the disease. To estimate the incubation period, click on the 'Incubation period' window in the application.

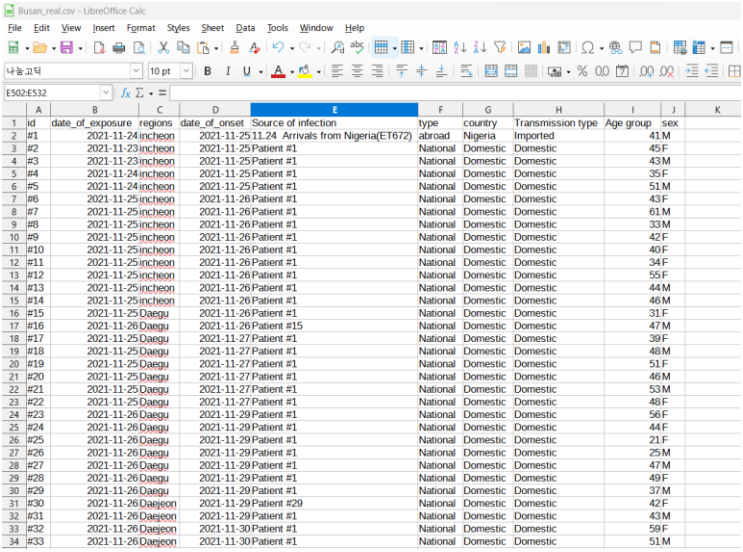
**Data preparation.**

1. Save your data spreadsheet in a '.csv' format example: "Busan_data.csv"

2. In the CSV data spreadsheet, the following variables should be present with the given specificities (Capital/small letter sensitive);

- ✓ *Date of exposure of the disease → rename as "date_of_exposure"
- ✓ *Date of symptom onset → rename as "date_of_onset"
- ✓ Gender/Sex → rename as "sex"
- ✓ Age/Age-group → rename as "age_group"
- ✓ Any other variable depending on the researchers' interests.

**Mock data for the COVID-19 outbreak.**



| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | id | date_of_exposure | regions | date_of_onset | Source of infection | type | country | Transmission type | Age group | sex |
| 2 | #1 | 2021-11-24 | incheon | 2021-11-25 | 11.24 Arrivals from Nigeria(ET672) | abroad | Nigeria | Imported | 41 | M |
| 3 | #2 | 2021-11-23 | incheon | 2021-11-25 | Patient #1 | National | Domestic | Domestic | 45 | F |
| 4 | #3 | 2021-11-23 | incheon | 2021-11-25 | Patient #1 | National | Domestic | Domestic | 43 | M |
| 5 | #4 | 2021-11-24 | incheon | 2021-11-25 | Patient #1 | National | Domestic | Domestic | 35 | F |
| 6 | #5 | 2021-11-24 | incheon | 2021-11-25 | Patient #1 | National | Domestic | Domestic | 51 | M |
| 7 | #6 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 43 | F |
| 8 | #7 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 61 | M |
| 9 | #8 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 33 | M |
| 10 | #9 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 42 | F |
| 11 | #10 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 40 | F |
| 12 | #11 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 34 | F |
| 13 | #12 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 55 | F |
| 14 | #13 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 44 | M |
| 15 | #14 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 46 | M |
| 16 | #15 | 2021-11-25 | Daegu | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 31 | F |
| 17 | #16 | 2021-11-26 | Daegu | 2021-11-26 | Patient #15 | National | Domestic | Domestic | 47 | M |
| 18 | #17 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 39 | F |
| 19 | #18 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 48 | M |
| 20 | #19 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 51 | F |
| 21 | #20 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 46 | M |
| 22 | #21 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 53 | M |
| 23 | #22 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 48 | F |
| 24 | #23 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 56 | F |
| 25 | #24 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 44 | F |
| 26 | #25 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 21 | F |
| 27 | #26 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 25 | M |
| 28 | #27 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 47 | M |
| 29 | #28 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 49 | F |
| 30 | #29 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 37 | M |
| 31 | #30 | 2021-11-26 | Daejeon | 2021-11-29 | Patient #29 | National | Domestic | Domestic | 42 | F |
| 32 | #31 | 2021-11-26 | Daejeon | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 43 | M |
| 33 | #32 | 2021-11-26 | Daejeon | 2021-11-30 | Patient #1 | National | Domestic | Domestic | 59 | F |
| 34 | #33 | 2021-11-26 | Daejeon | 2021-11-30 | Patient #1 | National | Domestic | Domestic | 51 | M |

This dataset can be downloaded from the ‘Download Mock data’ icon on the incubation period page.

**Required formats**

- ❖ date_of_exposure

  This is the probable date of exposure of the case, in the format “YYYY-MM-DD”

- ❖ date_of_onset

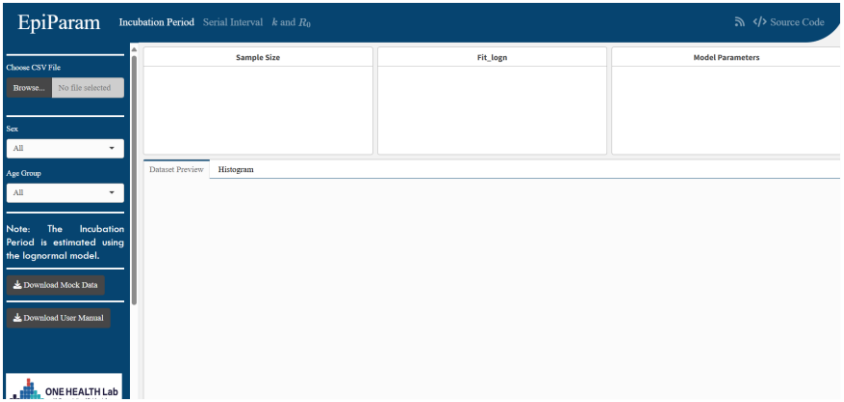
  This is the probable date of symptom onset of the disease, in the format “YYYY-MM-DD”

- ❖ sex

the gender of the individuals; All = Everyone, regardless of gender; Male = M; Female = F

- ❖ age

  The ages of the participants should be numeric in years. Classifications of the age into age include; All = Everyone regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥ 60

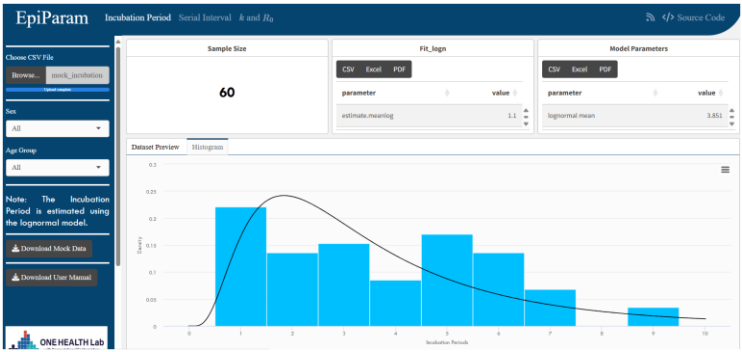



## Estimating the incubation period

To estimate the incubation period.

1. Click on the 'Incubation Period' icon at the top of the App.

2. Upload the .csv file containing the cleaned data you wish to use, by clicking on “Browse” and selecting the data from the computer.
3. This will produce an output as follows;




## Output Description

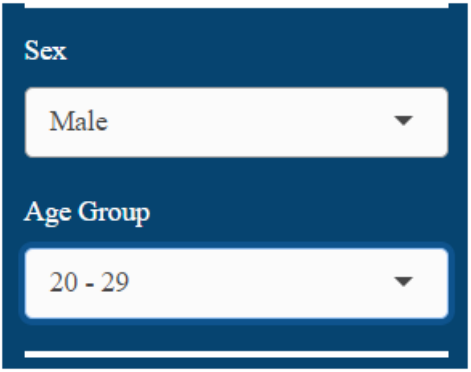
The output window is made up of 5 sub-windows.

- The “Sample Size” window, gives the total number of rows (individuals/patients/units) in the uploaded dataset.
- The Dataset Preview window, previews the uploaded dataset.
- The Fit_logn window that produces the model sub-parameters for the lognormal model.

- The Model Parameters window, provides estimates (mean, variance, 95th percentile using the log-normal model.
- Histogram window, demonstrates the output histogram of the distribution.

## Sub-group analysis

Based on individual needs, select stratifications by sex or by age group or a combination of both.




**Note:** EpiParam Version 1.0 uses the Lognormal distribution to estimate the incubation period. Future updates might include the Weibull and/or Gamma distributions!!!

## Serial interval (SI)

The serial interval is the time between successive cases in a disease chain of transmission. The serial interval is estimated from the interval between clinical onsets, in which case it is the 'clinical onset serial interval' when these quantities are observable.

**Data preparation.**

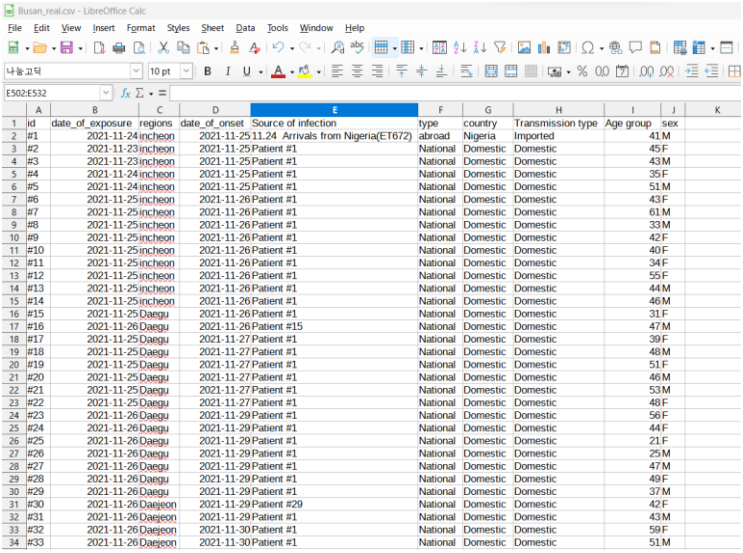
Similar to the incubation period;

- Save your data spreadsheet in a '.csv' format example: "Busan_data.csv"
- In the csv data spread sheet, the following variables should be present with the given specificities (Capital/small letter sensitive);

  - ✓ *Date of exposure of the disease → rename as "date_of_exposure"
  - ✓ *Date of symptom onset → rename as "date_of_onset"
  - ✓ *Source of infection → rename as "source_of_infection"

- ✓ Gender/Sex → rename as “sex”
- ✓ Age/Age-group → rename as “age_group”
- ✓ Any other variable depending on researchers’ interests

**Mock data for the COVID-19 outbreak.**



| id | date_of_exposure | regions | date_of_onset | Source of infection | type | country | Transmission type | Age group | sex |
|---|---|---|---|---|---|---|---|---|---|
| #1 | 2021-11-24 | incheon | 2021-11-25 | 11.24 Arrivals from Nigeria(ET672) | abroad | Nigeria | Imported | 41 | M |
| #2 | 2021-11-23 | incheon | 2021-11-25 | Patient #1 | National | Domestic | Domestic | 45 | F |
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| #10 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 40 | F |
| #11 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 34 | F |
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| #14 | 2021-11-25 | incheon | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 46 | M |
| #15 | 2021-11-25 | Daegu | 2021-11-26 | Patient #1 | National | Domestic | Domestic | 31 | F |
| #16 | 2021-11-26 | Daegu | 2021-11-26 | Patient #15 | National | Domestic | Domestic | 47 | M |
| #17 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 39 | F |
| #18 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 48 | M |
| #19 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 51 | F |
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| #21 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 53 | M |
| #22 | 2021-11-25 | Daegu | 2021-11-27 | Patient #1 | National | Domestic | Domestic | 48 | F |
| #23 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 56 | F |
| #24 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 44 | F |
| #25 | 2021-11-26 | Daegu | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 21 | F |
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| #31 | 2021-11-26 | Daejeon | 2021-11-29 | Patient #1 | National | Domestic | Domestic | 43 | M |
| #32 | 2021-11-26 | Daejeon | 2021-11-30 | Patient #1 | National | Domestic | Domestic | 59 | F |
| #33 | 2021-11-26 | Daejeon | 2021-11-30 | Patient #1 | National | Domestic | Domestic | 51 | M |

**Required formats**

- ❖ date_of_exposure

  This is the probable date of exposure of the case, in the format “YYYY-MM-DD”
- ❖ date_of_onset

This is the probable date of symptom onset of the disease, in the format "YYYY-MM-DD"

- id as id. A numeric identity which is given to each data unit
- source_of_infection

  The most probable source of infection of an individual given rise to the 'infector – infectee' pair
- sex

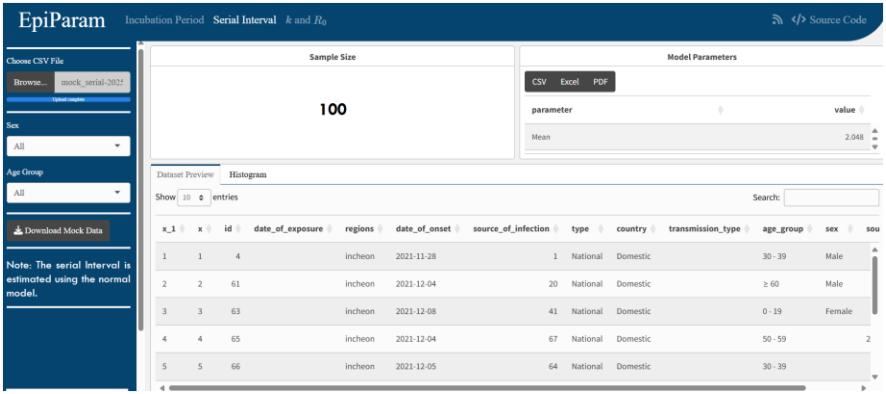
  the gender of the individuals; All = Everyone, regardless of gender; Male = Male; Female = Females
- age_group

  The ages of the participants in years and should be imputed as a whole number. Classifications of the age into age groups include; All = Everyone regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥ 60

**Estimating the Serial Interval**

In order to estimate the serial interval;

1. Click on the ‘serial interval’ icon at the top of the App.
2. Upload the .csv file containing the cleaned data you wish to use, by clicking on “Browse” and selecting the data from the computer.
3. This will produce an output as follows;




## Output Description (SI)

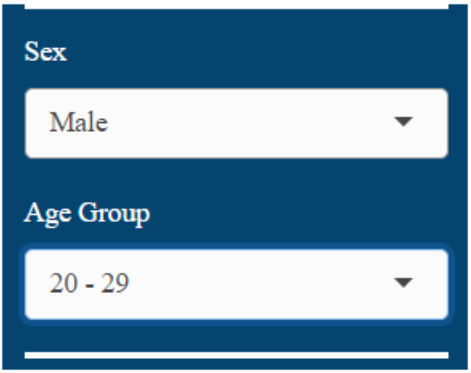
The output window is made up of 4 sub-windows.

- The “Sample Size” window, gives the total number of rows (individuals/patients/units) in the uploaded dataset.

- The Dataset Preview window previews the uploaded dataset.
- The Model Parameters window provides estimates (mean, variance, 25th,75th, and 95th quantile) using the normal model.
- The Histogram window demonstrates the output histogram of the distribution.

**Sub-group analysis**

Based on the researcher's needs, select stratifications by sex or by age group, or a combination of both.




**Note:** EpiParam Version 1.0 uses the normal distribution for the estimation of the serial interval!!!

## Kappa ($K$) and $R_0$

In the context of understanding and modelling transmission dynamics, the term "kappa" is sometimes used to represent the degree of variation in infectiousness among infected individuals. This variation in infectiousness is essential to understanding how different individuals may contribute differently to the spread of the disease.

### Interpretation

A higher kappa value suggests greater variation, meaning that some individuals are highly infectious (super spreaders), while others have lower infectiousness. A lower kappa value implies less variation, where most individuals have similar levels of infectiousness.

**Note:** The "kappa" and its associated values can vary depending on the modelling approach. The range of kappa values typically varies from 0 to 1, where:

**Kappa = 0** - Implies no variation in infectiousness among individuals. All infected individuals are equally infectious.

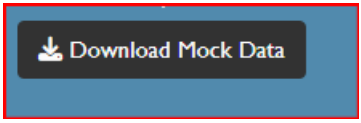
**Kappa = 1** - Indicates the highest level of variation in infectiousness, where there is extreme heterogeneity among infected individuals.

**Data preparation for parameter estimation**

3. Save your data spreadsheet in a '.csv' format
4. In the CSV data spreadsheet, the following variables should be present in your dataset;
    - ✓ Age - Age of the participants as whole numbers
    - ✓ Gender- Gender of the participants
    - ✓ Date of onset - Date of onset of symptoms for the respective disease
    - ✓ Date of confirm – Date of laboratory confirmation for the respective disease
    - ✓ Source of infection ID – The ID number of the individual from which the infection was gotten.
    - ✓ Cluster_id – Assigned ID of the infection cluster (A,B,C… etc or any other assigned style)

- ✓ Cluster character – Name or label that is given to the cluster
- ✓ Infectee generation – the number of successive generation of infections from a primary case in a specific cluster.

**Note:** A mock dataset has been provided for your information. You can download the mock data in the 'Download Mock data' tab in the app.



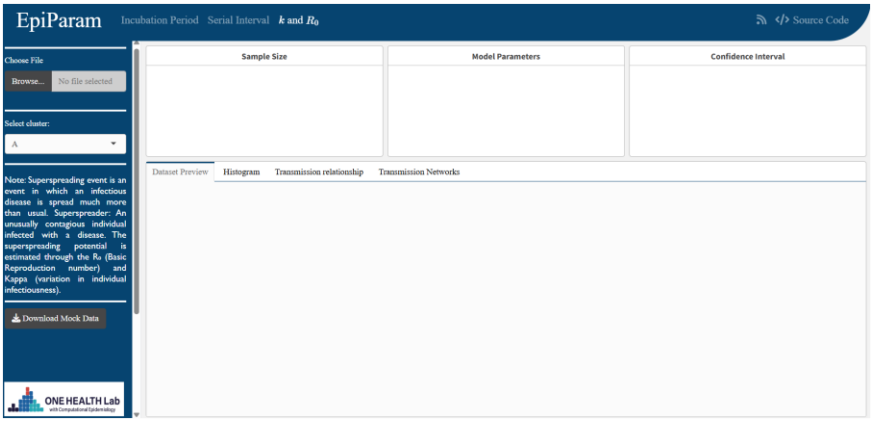

**Required formats**

- ❖ Date of confirm – Date of laboratory confirmation in the format "DD -MM-YYYY"
- ❖ Date of onset - This is the probable date of symptom onset of the disease, in the format "DD -MM-YYYY"
- ❖ Gender

  The gender of the individuals; All = Everyone, regardless of gender; Male = Male; Female = Females

❖ Age

The ages of the participants in years and should be imputed as a whole number. Classifications of the age into age groups include; All = Everyone regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥60

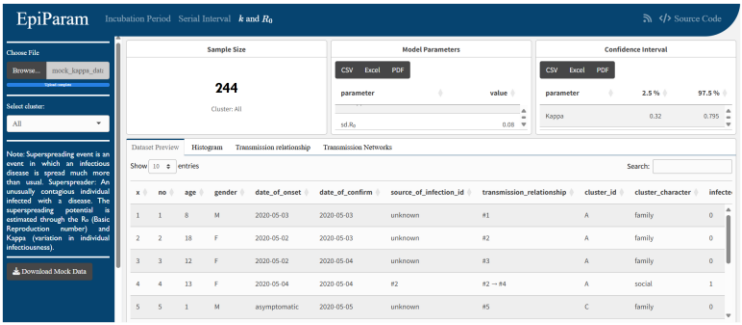



**Estimating the Kappa (*K*) and $R_0$**

To estimate the K and $R_0$;

4. Click on the 'K and $R_0$' icon at the top of the App.
5. Upload the .csv file containing the cleaned data you wish to use, by clicking on "Browse" and selecting the data from your device.

6. This will produce an output as follows;

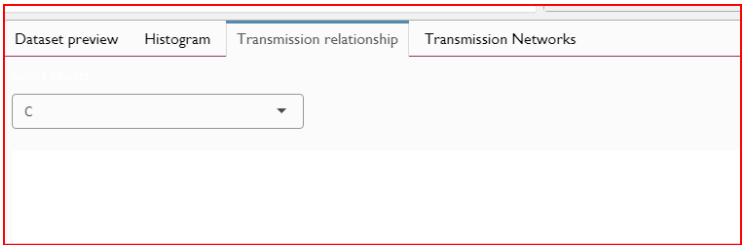



## Output Description

The output window is made up of 7 sub-windows.

- The 'Sample Size' window, gives the total number of rows (individuals/patients/units) in the uploaded dataset.
- The 'Dataset preview' window previews the uploaded dataset.
- The 'Model parameter' window that produces estimated K and $R_0$ values as well as the corresponding standard deviations (SD).
- The 'Confidence Interval' window provides the confidence interval of each of the parameters.

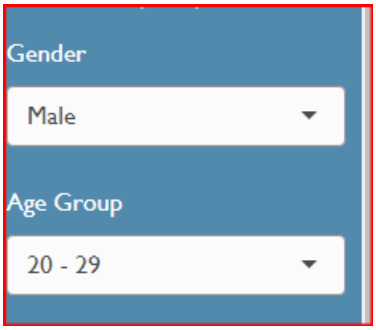
- ‘Histogram window’ demonstrates the output histogram of the distribution.
- The ‘Transmission relationship’ window demonstrates the transmission relationship of each cluster. This can be viewed by selecting the cluster ID in the dropdown section of the transmission relationship window, as shown below;




- The ‘Transmission networks’ window demonstrates the network connections among the infector and the infectee.

**Sub-group analysis**

Based on individual needs, select stratifications by sex or by age group or a combination of both.


Gender
Male
Age Group
20 - 29